

Oracle EPM Predictive Cash Forecasting

Cash Management Integration - Advanced Lab Guide

Product Design Documentation • Implementation Guide • Training Material • Technical Reference • Troubleshooting Handbook

Advanced-stage focus

Integrate Oracle Fusion Cash Management and bank-position data into Predictive Cash Forecasting to establish opening cash, monitor available/booked balances, reconcile bank accounts, calculate closing cash, control liquidity thresholds, automate loads, and troubleshoot production failures.

Lab Property	Design
Primary horizon	13-week weekly rolling liquidity model
Primary source	Oracle Fusion Cash Management + bank balance / statement data
Opening cash basis	Closing Available or Closing Booked Balance, per policy
Core outputs	Opening Cash, Closing Cash, Cash Buffer, Cash Over Buffer / headroom
Audience	Advanced EPM consultants, treasury, integration architects, cash managers

1. Purpose and Learning Outcomes

Why this integration matters

Cash Management integration anchors a predictive forecast to actual liquidity. If opening cash is stale, duplicated, mapped to the wrong entity, or inconsistent across bank accounts, every downstream forecast and funding decision becomes unreliable.

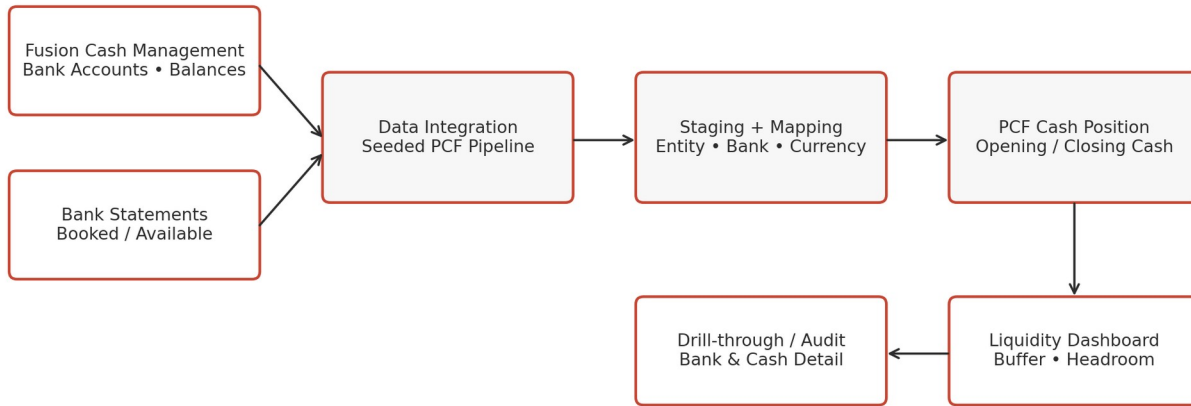
- Design the end-to-end Fusion Cash Management to EPM PCF integration pattern.
- Choose and govern opening-cash source logic using available vs booked balances.
- Map bank accounts, entities, currencies and reporting intersections.
- Reconcile bank-source totals to loaded PCF actuals and cash positions.
- Detect stale balances, duplicate bank/date records and missing account mappings.
- Calculate closing cash and liquidity headroom across scenarios.
- Automate seeded pipelines, validation gates, retries and controlled restarts.
- Implement drill-through, security, audit and troubleshooting procedures.

Design Principles

Principle	Implementation meaning
Source-of-truth first	Use authoritative bank/cash data as the starting point for liquidity.
Freshness control	Reject or flag stale cash data before publish.
Explainability	Every opening balance must be traceable to bank/account/date/currency.
Exact reconciliation	Bank/cash control totals must reconcile before forecast publication.
Policy-driven selection	Available vs booked balance is a documented business decision, not an ad hoc choice.

2. Reference Architecture

Cash Management Integration Reference Architecture



Primary design goal: convert trusted bank and cash-position data into explainable opening/closing cash and liquidity visibility.

The reference architecture separates source extraction, staging/mapping, PCF target loading and liquidity calculation. The design supports operational bank-level traceability while publishing controlled cash-position totals for forecasting and treasury decisioning.

Key Source Components

Source object	Examples	PCF use
Bank account	Bank, branch, account, legal entity, currency	Bank-level analysis and target mapping
Bank balance	Available, booked/ledger, balance date/time	Opening cash position
Bank statement	Statement date, closing balance, transaction lines	Reconciliation and source evidence
Cash transactions	Receipts, payments, transfers	Actual cash-movement visibility
External bank/TMS feed	Intraday or end-of-day positions	Supplement or cross-check Fusion data

3. Cash Position Functional Model

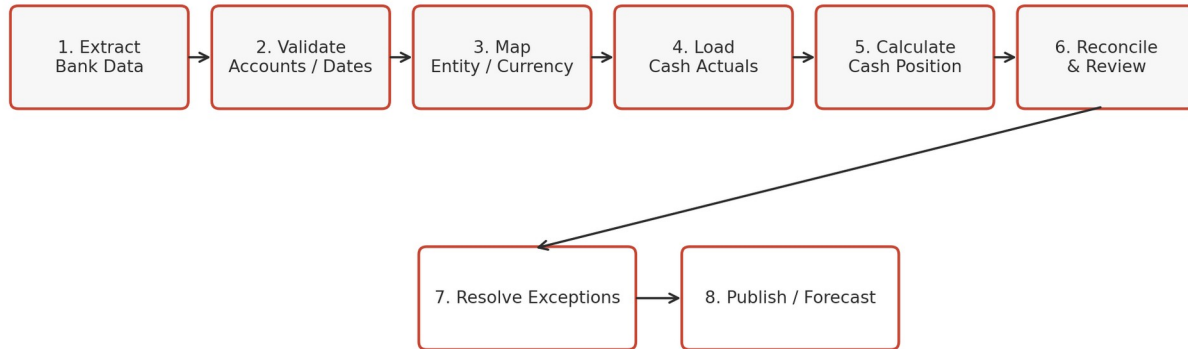
Line / Measure	Definition	Typical derivation
Opening Cash Balance	Starting liquidity for forecast period	Selected closing available/booked balance from prior cut-off
Operating Cash Flow	AR receipts less AP/payroll/tax/opex	Forecast methods and transactional sources
Investing Cash Flow	Capex, disposals, investments	Investing forecast
Financing Cash Flow	Debt, interest, dividends, equity	Financing forecast
Net Cash Flow	Operating + Investing + Financing	Calculated
Closing Cash Balance	Opening Cash + Net Cash Flow	Calculated
Cash Buffer	Minimum required liquidity	Policy / assumption
Cash Over Buffer	Closing Cash - Cash Buffer	Calculated liquidity headroom

Core control equation

Closing Cash = Opening Cash + Operating Cash Flow + Investing Cash Flow + Financing Cash Flow. Cash Over Buffer = Closing Cash - Cash Buffer.

4. Cash Management Integration Process

Cash Management Integration Process Flow



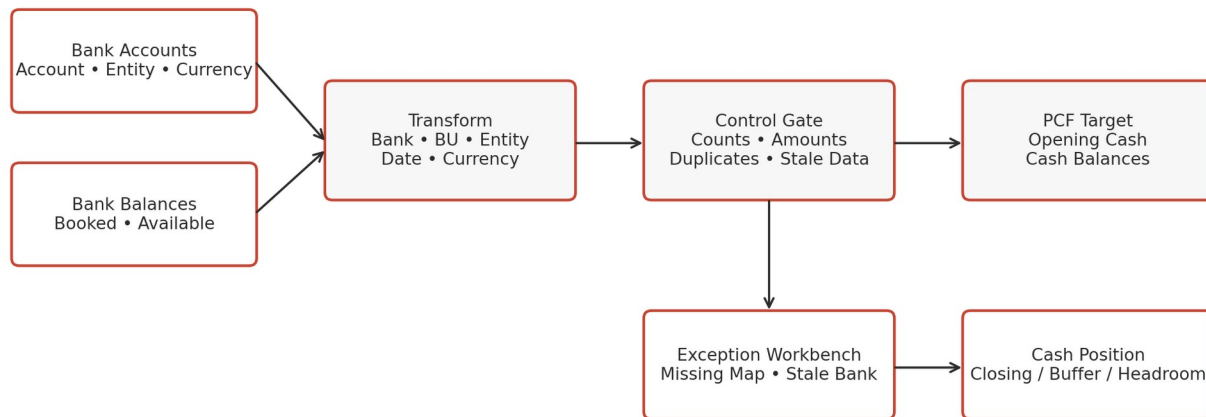
1. Extract bank accounts, bank balances, statement and cash-position data from Fusion Cash Management.
2. Validate account status, dates, currencies, legal-entity ownership and freshness.
3. Map source bank/account/business-unit information to EPM Entity, Bank and custom dimensions.
4. Load cash actuals and selected opening-balance source into PCF.
5. Calculate opening cash, net cash movement and closing cash.
6. Reconcile source balances and investigate exceptions.
7. Resolve stale, duplicated, unmapped or out-of-balance records.
8. Publish trusted cash position to the rolling forecast and liquidity dashboard.

Daily/Weekly Operating Cadence

Timing	Activity	Owner
Early AM	Load bank balances / Cash Management data	Integration Admin
After load	Reconcile accounts and source totals	Cash Manager
Forecast run	Calculate opening/closing cash and scenarios	Treasury
Exception window	Resolve stale/missing/unmapped items	Cash Ops + IT
Approval	Approve liquidity position and publish	Treasurer / Controller

5. Data Flow and Control Gates

Cash Management Data Flow and Control Gates



Mapping Framework

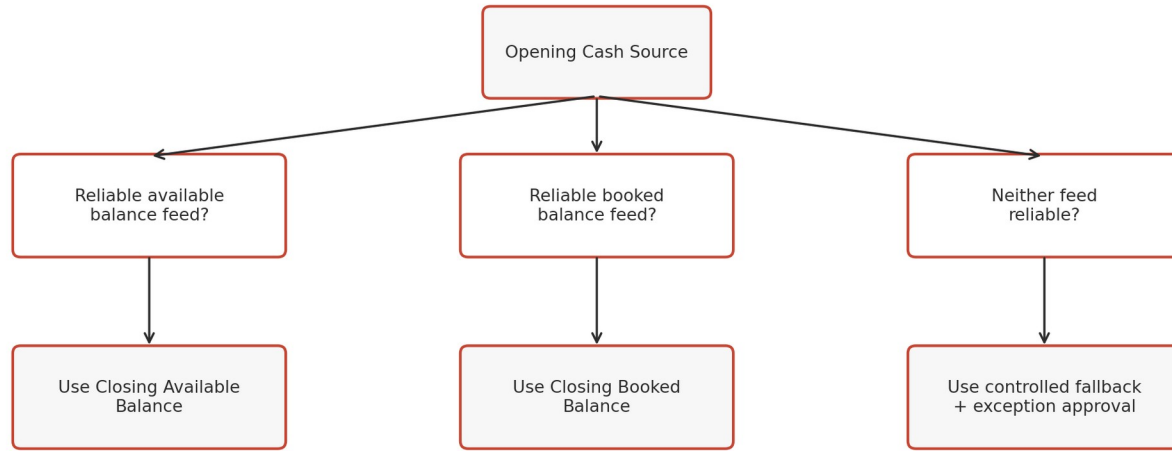
Source attribute	Target / purpose	Control
Bank Account	Bank Account custom dimension or analytical attribute	No unmapped active account
Legal Entity / BU	Entity	One authoritative mapping
Currency	Currency / reporting translation	Currency must exist
Balance Date	Period / As-of date	No future/stale date beyond policy
Balance Type	Available vs Booked	Allowed values only
Amount	Opening Cash input	Duplicate and sign validation

Control gate

A successful technical load is not enough. Validate account count, amount by currency/entity, data freshness, duplicate keys, rejected rows, and source-to-target financial reconciliation before publishing cash.

6. Opening Balance Source Decision Tree

Opening Cash Balance Selection Decision Tree



Policy Guidance

Situation	Recommended basis	Reason
Available balance is timely and trusted	Closing Available Balance	Closer to immediately usable liquidity
Available balance unavailable but booked is trusted	Closing Booked Balance	Stable accounting/bank position basis
Both feeds differ materially	Exception review before publish	Investigate holds, float, uncleared items, timing
No reliable bank balance	Controlled fallback + approval	Avoid silent carry-forward of stale cash

7. Oracle-Inspired Configuration Mockup

Oracle-Inspired Cash Management Configuration Mockup

Predictive Cash Forecasting > Administrator > Data Integration > Cash Management

Source
Oracle Fusion Cash Management

Opening Balance
Closing Available Balance

Bank Dimension
Bank Account

Currency
Entity Currency

Staleness Threshold
24 hours

Duplicate Policy
Reject + Log

Reconciliation
Exact control total

Save & Validate

1. Configure the Fusion Cash Management connection and integration scope.
2. Choose opening-cash source policy: Closing Available or Closing Booked balance.
3. Configure Bank Account and Entity mappings.
4. Set freshness threshold and duplicate-handling rules.
5. Set reconciliation tolerance and exception severity.
6. Test one entity and small bank-account population before enterprise rollout.

Mockup note

The UI in this guide is Oracle-inspired for training and design discussion; it is not a literal reproduction of Oracle proprietary screens.

8. Advanced Lab Scenario - Cash Position Integration

Company: BISP Manufacturing Group | As-of: 31-Aug-2026 | Reporting Currency: USD | Minimum Cash Buffer: \$6.5M

Bank Account	Entity	Currency	Booked (\$M)	Available (\$M)	Status
Operating USD	US01	USD	6.20	5.90	Fresh
Payroll USD	US01	USD	2.10	2.10	Fresh
Tax USD	US01	USD	1.40	1.35	Fresh
EU Operating	EU01	EUR	2.50	2.20	Stale 30h
India Operating	IN01	INR	82.0	80.5	Fresh

Lab policy

Use Closing Available Balance as opening cash. A balance older than 24 hours is a WARNING and cannot be published without reviewer acknowledgement or an approved fallback.

9. Lab Step 1 - Validate and Reconcile Bank Data

Control	Expected	Observed	Result
US active bank accounts	3	3	PASS
US available balance	9.35M	9.35M	PASS
EU feed freshness	<=24h	30h	WARNING
Duplicate bank/date/currency	0	0	PASS
Unmapped accounts	0	0	PASS

EU exception

EU01 cannot use its stale available balance without review. Management may use the booked balance, a prior approved balance, or wait for a refreshed feed depending on policy.

10. Lab Step 2 - Calculate Opening and Closing Cash

For US01, opening available cash = \$5.90M + \$2.10M + \$1.35M = \$9.35M.

Week	Opening Cash	Operating CF	Investing CF	Financing CF	Closing Cash	Headroom
W1	9.35	+1.20	-0.90	-0.40	9.25	2.75
W2	9.25	+0.80	-1.10	-0.20	8.75	2.25
W3	8.75	-0.30	-0.50	+0.00	7.95	1.45
W4	7.95	-1.10	-0.60	+0.50	6.75	0.25

All amounts \$M. Minimum cash buffer = \$6.50M.

Treasury insight

W4 remains above the buffer by only \$0.25M. Any negative bank-position adjustment or forecast miss could create a breach; Treasury should prepare a management-action scenario.

11. Stress Scenario and Funding Action

Stress: a \$1.0M receipt slips from W4 to W5 and EU available cash is \$0.4M below booked cash after translation.

Scenario	W4 Closing	Buffer Headroom	Action
Base	6.75	+0.25	Monitor
Stress	5.75	-0.75	Funding or payment timing action required
Management Action	6.85	+0.35	Draw \$1.10M revolver or accelerate collections

Decision surface

Cash Management data turns scenario forecasts into a funding decision by anchoring the forecast to trusted opening liquidity and exposing the buffer impact.

12. Available vs Booked Balance Reconciliation

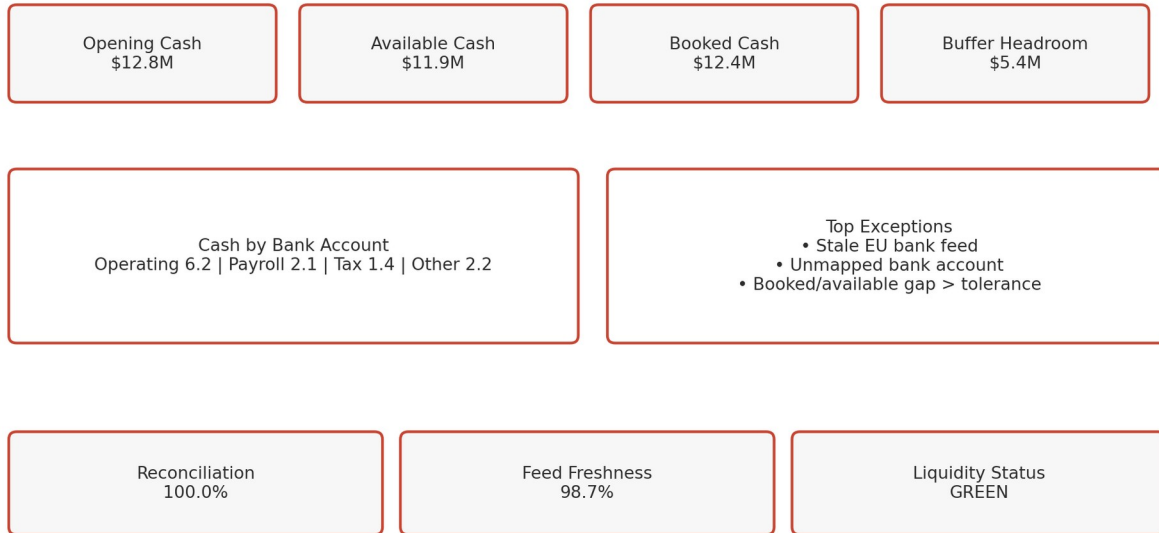
Cause of difference	Available impact	Booked impact	Treatment
Uncleared deposit	May be excluded	May be included later	Investigate timing
Payment hold / reserved cash	Reduces availability	Booked may remain higher	Use available for liquidity if policy supports
Bank float	Timing difference	Statement/accounting basis differs	Reconcile daily
Intraday transaction	Available changes first	Booked may lag	Use cut-off policy
Data staleness	Either can be wrong	Either can be wrong	Block or flag based on freshness

Governance

Do not switch between available and booked balances opportunistically to improve liquidity. The balance basis must be documented and consistently applied.

13. Dashboard and Liquidity Monitoring

Cash Management Integration Dashboard Mockup



Recommended KPIs

KPI	Interpretation
Opening Cash	Trusted starting cash by entity / bank / currency
Available vs Booked Gap	Liquidity availability / timing difference
Reconciliation %	Source control total match
Feed Freshness	Percentage of accounts inside freshness SLA
Cash Buffer Headroom	Closing cash less minimum buffer
Stale Accounts	Bank positions exceeding freshness threshold
Unmapped Bank Amount	Cash not assigned to valid target intersection

14. Fusion Cash Management Integration Design

1. Configure Cloud EPM to Oracle Cloud ERP connection and access.
2. Use the ready-to-use Cash Management integration / seeded pipeline for PCF where applicable.
3. Load bank account and balance data into the configured PCF target.
4. Reconcile account count and amounts by entity/currency.
5. Use transaction or analytical drill-through where supported to investigate cash-position detail.
6. Publish only after validation and approval controls pass.

Security

Separate integration administration, cash review, forecast approval and audit responsibilities. Bank-account visibility should align with entity/business scope and treasury policy.

15. Automation and Pipeline Orchestration

Seq	Job	Failure policy
1	Extract Cash Management data	Stop on source failure
2	Validate bank/account metadata	Stop on active unmapped account
3	Load balances and cash data	Reject duplicate keys
4	Reconcile control totals	Block publish on tolerance breach
5	Calculate opening/closing cash	Restart from calculation after correction
6	Refresh reporting/dashboard	Warning if delayed
7	Publish Approved cash position	Approval required

Restartability

- Persist run ID, source job ID, account count and totals.
- Use scoped clear/reload or idempotent loading to prevent double counting.
- Keep rejected records and exception evidence.
- Restart from the failed logical stage, not the entire chain, when safe.

16. Security, Workflow and Audit

Role	Responsibility
Service Administrator	Connections, pipelines, metadata, technical configuration
Cash Manager	Bank-position review, reconciliation, cash assumptions
Treasury Analyst	Liquidity analysis and scenario actions
Controller / Reviewer	Reconciliation and override approval
Treasurer	Funding and liquidity decisions
Auditor	Read-only evidence, run logs, balance policy and approvals

Audit fields

Run ID • Source account • Balance type • As-of date/time • Source amount • Target amount • Old/New override value • Reason • User • Approval status.

17. Log / Error Examples

Cash Management Log / Error Examples

ERROR CM-204 Bank account 001-998 has no Entity mapping. Record rejected.

WARN CM-118 Available balance feed is 30 hours old; threshold is 24 hours.

ERROR CM-312 Duplicate bank/date/currency balance detected. Load blocked.

INFO CM-001 86 bank accounts loaded; source and target control totals reconciled.

Troubleshooting Matrix

Symptom	Likely cause	Resolution
Opening cash too high	Duplicate account/date balance	Reject duplicate key; reload scope
Opening cash too low	Missing bank/account mapping	Correct mapping and reconcile
Balance stale	Upstream bank/Cash Management timing	Refresh feed or approved fallback
Wrong entity	Bank ownership mapping incorrect	Fix Entity mapping
Available/booked mismatch	Float, holds, uncleared items	Reconcile by bank/account
Pipeline green but cash wrong	No financial control total gate	Add exact reconciliation before publish
Drill-through denied	ERP/EPM access mismatch	Align roles and data access

18. Performance and Production Controls

- Keep bank-account granularity where needed for reconciliation, but avoid unnecessary high-cardinality custom dimensions.
- Define cut-off times by bank/region and document timezone handling.
- Track freshness SLA, reconciliation %, rejected amount, duplicate count and pipeline duration.
- Use entity/bank partitioning only when it materially improves restartability or volume handling.
- Archive balance snapshots to support audit and forecast-vs-actual analysis.
- Test month-end, quarter-end and high-volume bank-statement conditions.

Production KPI	Example target
Fresh balance coverage	>= 99% of material accounts within SLA
Unmapped active bank amount	0
Duplicate balance records	0
Financial reconciliation	100% or approved tolerance
Critical incident response	Within 15 minutes
Cash position publish	Before treasury review cut-off

19. Forecast vs Actual and Cash-Position Diagnostics

Diagnostic	Purpose
Opening Cash variance	Detect late/stale/incorrect balance source
Net Cash Flow variance	Separate forecast-method error from opening-cash error
Closing Cash variance	End-to-end liquidity accuracy
Available vs Booked delta	Operational liquidity vs accounting timing
Forecast Value Add	Determine whether scenario/override improved decision quality

Important distinction

A forecast can be mathematically correct but operationally wrong if the opening cash position is stale. Diagnose opening-cash accuracy separately from forecast-flow accuracy.

20. Implementation Checklist

- ☐ Define opening balance source policy (Available or Booked).
- ☐ Inventory all material bank accounts and ownership.
- ☐ Configure Fusion Cash Management connection and integration.
- ☐ Build bank/account/entity/currency mappings.
- ☐ Set freshness and duplicate rules.
- ☐ Define reconciliation control totals and tolerances.
- ☐ Configure cash buffer by entity / group.
- ☐ Build Base, Stress and Management Action scenarios.
- ☐ Configure approval and override evidence.
- ☐ Test stale, duplicate, missing-map and wrong-currency exceptions.
- ☐ Test restart behavior and duplicate prevention.
- ☐ Build dashboard and liquidity exception views.
- ☐ Complete security/SoD review.
- ☐ Execute UAT with Treasury, Cash Ops, Controller and IT.
- ☐ Prepare production runbook and escalation matrix.

21. Advanced Student Challenge

Scenario

A \$4.0M bank account shows Booked Balance \$4.0M but Available Balance \$3.1M because \$0.9M is reserved for same-day outgoing payments. The group forecast shows a \$0.4M buffer breach if Available is used, but no breach if Booked is used.

- Choose the appropriate opening-balance basis and justify it.
- Explain why switching to Booked merely to avoid a breach is a governance failure.
- Design the exception/approval workflow if the available balance feed is late.
- Show the impact on closing cash, funding requirement and liquidity headroom.
- List the source evidence an auditor should be able to trace.

Consultant-level takeaway

Cash Management integration is not just a data load. It is the control layer that determines whether the forecast begins from real, usable liquidity and whether treasury decisions are defensible.

22. Technical Reference and Completion Criteria

Capability	Cash Management relevance
Fusion ERP integration	Provides ready-to-use PCF integration patterns for Cash Management data.
Opening Cash Balance	Can be sourced using supported closing balance concepts, governed by application configuration/policy.
Bank / cash traceability	Supports source-to-target reconciliation and operational analysis.
Data Integration pipelines	Orchestrate source loads and downstream processing.
Cash Buffer / Cash Over Buffer	Provides liquidity threshold and headroom visibility.
Rolling forecast	Uses trusted opening cash plus projected cash flows to produce closing cash.

- All material bank accounts are mapped and reconciled.
- Opening cash source basis is documented and consistently applied.
- Freshness, duplicate and reconciliation controls block bad data.
- Cash position calculation and buffer headroom are validated.
- Automation can restart safely without duplicate loading.
- Exceptions and overrides retain evidence and approvals.
- The dashboard surfaces stale feeds, cash gaps and liquidity breaches.
- The runbook defines ownership, SLA, escalation and recovery.